Ch. 10.2: Properties of Power Series

Calculus III | MTH 331 | Section A | Fall 2012

Dr. James Ashe

Johnson C. Smith University

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Definition

A *power series* is an infinite series of the form

$$\sum_{k=0}^{\infty} c_k x^k = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + c_{n+1} x^{n+1} \ldots$$

or, more generally,

$$\sum_{k=0}^{\infty} c_k (x-a)^k = c_0 + c_1(x-a) + \cdots + c_n(x-a)^n + \ldots$$

where c_k are constants.

- The constant a is the **center**.
- The set of x's such that the series converges is called the **interval of convergence**, $I = [c,d]$. i.e., $I = \{x \in \mathbb{R} | \sum_{k=0}^{\infty} c_k (x-a)^k < \infty\}$.
- The **radius of convergence** of the series, R, is the distance from a to d, i.e., $R = |d-a|$.

Finding an interval of convergence

ex. $\sum \frac{x^k}{2^k}$

Need to find all x such that the series converges. By the Ratio Test, $\lim_{k\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{k\to\infty} \left|\frac{x^{k+1}/2^{k+1}}{x^k/2^k}\right| = \lim_{k\to\infty} \frac{|x|}{2} = \frac{|x|}{2}$

So this series diverges when $\frac{|x|}{2} > 1 \Rightarrow x < -2$ and $x > 2$.

- Need to test the endpoints.

At $x=-2$, $\sum_{k=0}^{\infty} \frac{(-2)^k}{2^k} = 1 - 1 + 1 - 1 + \cdots$, so it diverges.

At $x = 2$, $\sum_{k=0}^{\infty} \frac{(2)^k}{2^k} = 1 + 1 + \cdots$, so it diverges.

So the interval of convergence is $-2 < x < 2$.

ex. $\sum_{k=1}^{\infty} \frac{(x-2)^k}{nk^n}$

Using the Ratio Test,

$$\lim_{k\to\infty} \left(\frac{n}{n+1} \cdot \frac{5^n}{5^{n+1}} \cdot \frac{|x-2|^{n+1}}{|x-2|^n}\right) = \frac{1}{5}|x-2|.$$

$\frac{1}{5}|x-2| > 1 \Rightarrow |x-2| > 5$, so then the series diverges when $x \notin (-3, 7)$.

For $x = 7$, the series is $\sum_{k=1}^{\infty} \frac{1}{k}$ is harmonic so it diverges.

For $x = -3$, the seres is $\sum_{k=1}^{\infty} (-1)^k \frac{1}{k}$, which is the alternating harmonic series which converges (by the Alternating Series Test).

So the interval of convergence is $-3 \leq x < 7$.

Finding a power series representation for $f(x)$ and its interval of convergence

$$g(x) = \frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \text{ when } |x| < 1.$$

- The geometric series, $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$ is a power series (replacing the r with a variable; each $c_i = 1$).

ex. Find a power series for $f(x) = \frac{1}{1+x^3}$ centered at 0.

Substitute $-x^3$ for x in $g(x)$. This gives $\frac{1}{1-(-x^3)}$, and

$\frac{1}{1-(-x^3)} = \sum_{k=0}^{\infty} (-x^3)^n$ *provided* that $|-x^3| < 1$.

Now $\sum_{k=0}^{\infty} (-x^3)^n = \sum_{k=0}^{\infty} (-1)^n x^{3n}$ and $|x^3| < 1 \Rightarrow |x| < 1$.

ex. 33 Find the power series for $h(x) = \frac{1}{(1-x)^2}$ using $g(x) = \frac{1}{1-x}$.

Note: $\frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right)$, and

$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ when $|x| < 1$.

So then $h(x) = \frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\sum_{k=0}^{\infty} x^k = \sum_{k=1}^{\infty} kx^{k-1}$.

Homework due (10/24)
10.2: 9-12, 21-24, 34-35, 39-41